CNNs Are Still Worth Understanding

ARTIFICIAL INTELLIGENCE · MACHINE LEARNING · DEEP LEARNING

CNNs Are Still Worth Understanding

SWIPE ->

Before jumping straight into vision transformers, it's worth understanding why CNNs became so important.

A Convolutional Neural Network doesn't process an image as one giant, unstructured collection of pixels.

It learns filters that respond to local visual patterns.

Early layers may detect simple edges and textures.

Deeper layers can combine those patterns into more complex structures.

A simplified CNN pipeline looks like this:

Image Convolution Activation Downsampling Deeper features Prediction

Three ideas make CNNs particularly useful:

- Local receptive fields help capture nearby visual patterns.
- Shared convolutional weights reduce the number of parameters compared with fully connected image processing.
- Hierarchical features allow the network to build complex representations from simpler ones.

This doesn't mean CNNs understand images exactly like humans do.

They learn statistical representations from data, and their performance depends on the architecture, training process, and distribution of the inputs.

But understanding CNNs makes it much easier to understand architectures such as VGGNet, ResNet, DenseNet, and MobileNet.

The best way to learn modern computer vision isn't to memorize model names.

It's to understand the architectural problem each model was trying to solve.

#CNN #DeepLearning #ComputerVision

Thanks for reading.

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Code: github.com/yasirusman85/Tank